

Abstract: Science and Engineering Fair of Houston

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Utilizing Grad-CAM Based Training Loss and Self-Attention for Kidney Cancer

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Category:

Software Design

Clinical adoption of AI is limited by one ethical problem: high accuracy is not enough if physicians cannot tell whether a model is using medically valid evidence. Popular explainability methods like Grad-CAM, Guided Grad-CAM, and Grad-CAM++ generate heatmaps after training, but they do not influence what the network learns. This leaves a safety gap where a system can appear reliable while depending on external cues. This project uses Tensorflow and introduces a kidney tumor detector based on images from CT KIDNEY DATASET: Normal-Cyst-Tumor and Stone. Instead of adding interpretation later, we calculate Grad-CAM heatmaps during every training iteration and use them as part of the loss function. The model receives gradient feedback not only for classification accuracy but also for the quality of its explanations. We penalize scattered attention patterns and reward focused activations through three complementary metrics: We add an entropy term to the loss that encourages the model to concentrate attention on specific regions rather than distributing it across the entire image. We measure the spatial extent of the heatmap and penalize models that activate large portions of the image, forcing more selective feature usage. We compute the variance-to-mean ratio of activation intensities, rewarding models that produce sharp peaks of attention rather than weak activations. A model trained with tumor-focused Grad-CAM loss cannot "cheat", it must learn genuine tumor morphology, texture, and boundary characteristics to minimize its loss. This creates inherently safer, more generalizable models suitable for real-world clinical deployment across different hospitals, scanners, and patient populations.

1. As a part of this research project, the student directly handled, manipulated, or interacted with (check all that apply):

- ☐ Human participants ☐ potentially hazardous biological agents
☐ Vertebrate animals ☐ microorganisms ☐ rDNA ☐ tissue

2. This abstract describes only procedures performed by me/us, reflects my/our own independent research, and represents one year's work only.

- ☒ yes ☐ no

3. I/We worked or used equipment in a regulated research institution or industrial setting.

- ☐ yes ☒ no

4. This project is a continuation of previous research.

- ☐ yes ☒ no

5. My display board includes non-published photographs/visual depictions of humans (other than myself):

- ☐ yes ☒ no

6. I/We hereby certify that the abstract and responses to the above statements are correct and properly reflect my/our own work.

- ☒ yes ☐ no

